

Q.No 7:

Question Description:

Simon is studying B.Tech.-Mechanical Engineering.

He's going to attend a computer science-based subject exam this semester.

Due to the less preparation in the previous monthly tests, his internal mark decreased.

His computer science Professor made an offer one more chance to boost up his internal marks.

Professor assigns a program to Simon for the internal mark bootup.

So Simon wants to solve the given task is, Given two arrays, A and B, of equal size n,

the task is to find the minimum value of $A[0] * B[0] + A[1] * B[1] + \dots + A[n-1] * B[n-1]$,

where shuffling of elements of arrays A and B is allowed.

can you help him in solving Questions ?

Constraints:

$1 \leq T \leq 100$

$1 \leq N \leq 50$

$1 \leq A[i] \leq 20$

Input Format:

The first line of input contains an integer denoting the no of test cases.

Then T test cases follow. Each test case contains three lines.

The first line of input contains an integer N denoting the size of the arrays.

In the second line are N space separated values of the array A[], and in the last line are N space separated values of the array B[].

Output Format:

For each test case in a new line print the required result.

Example :

Input : A[] = { 3, 1, 1 } and B[] = { 6, 5, 4 }.

Output : 23 Minimum value of $S = 1 * 6 + 1 * 5 + 3 * 4 = 23$.

Input : A[] = { 6, 1, 9, 5, 4 } and B[] = { 3, 4, 8, 2, 4 }

Output : 80. Minimum value of $S = 1 * 8 + 4 * 4 + 5 * 4 + 6 * 3 + 9 * 2 = 80$.

Logical Test Cases

Test Case 1

INPUT (STDIN)

```
3
4
2 5 3 7
8 5 7 3
3
13 11 1
6 15 14
```

Test Case 2

INPUT (STDIN)

```
2
3
13 11 1
6 15 14
5
6 11 9 15 4
3 41 8 21 4
```

Code:

```
#include <stdio.h>

void sort(int a[], int n, int flag)
{
```

```
int i, j, temp;

for (i = 0; i < n - 1; i++)
{
    for (j = i + 1; j < n; j++)
    {
        if (flag == 0)
        {
            if (a[i] > a[j])
            {
                temp = a[i];
                a[i] = a[j];
                a[j] = temp;
            }
        }
        else
        {
            if (a[i] < a[j])
            {
                temp = a[i];
                a[i] = a[j];
                a[j] = temp;
            }
        }
    }
}
```

```
int main()
{
    int T;
    scanf("%d", &T);

    int A[50], B[50];
    int answer[100];

    for (int t = 0; t < T; t++)
    {
        int n;

        scanf("%d", &n);
        for (int i = 0; i < n; i++)
        {
            scanf("%d", &A[i]);
        }
        for (int i = 0; i < n; i++)
        {
```

```
scanf("%d", &B[i]);  
}  
  
\  
sort(A, n, 0);  
\  
sort(B, n, 1);  
  
int sum = 0;  
  
for (int i = 0; i < n; i++)  
{  
    sum = sum + A[i] * B[i];  
}  
  
answer[t] = sum;  
}  
  
printf("Output:\n");  
  
for (int t = 0; t < T; t++)  
{  
    printf("%d\n", answer[t]);  
}  
  
return 0;  
}
```

Output:

Test_case 1:

```
3
4
2 5 3 7
8 5 7 3
3
13 11 1
6 15 14
5
6 11 9 15 4
3 41 8 21 4
Output:
83
247
451
```

Test_case 2:

```
2
3
13 11 1
6 15 14
5
6 11 9 15 4
3 41 8 21 4
Output:
247
451
```

Q.no 8:

Question description

kkalaiselvan is going to behave as a vehicle driver.

he needs to drive an automobile on a track divided into "N" no. of sub-tracks.

The letter "K" is also assigned to you.

i.e. the maximum number of kilometres a vehicle may go on each sub-track.

You can add any unit of Petrol to the automobile if it can't cover a sub-track. The total kilometres your automobile may go increases by one unit for every unit of gasoline added.

Input:

The first line of input contains an integer T denoting the no of test cases. Then T test cases follow. Each test case contains two space separated integers N and K.

The second line of each test case contains N space separated integers (A[]) denoting the distance of each N sub-tracks.

Output:

For each test case in a new line you have to print out the minimum unit of Petrol your car require to cover all the sub-tracks. If no extra unit of petrol is required, print -1.

✓ Logical Test Cases

Test Case 1

INPUT (STDIN)

```
2
5 6
2 5 4 5 2
5 3
1 6 3 5 2
```

EXPECTED OUTPUT

```
-1
3
```

Test Case 2

INPUT (STDIN)

```
2
5 6
12 15 14 15 12
5 3
11 61 31 51 21
```

EXPECTED OUTPUT

```
9
58
```

✓ Mandatory Test Cases

Code:

```
#include <stdio.h>

void sort(int a[], int n)
{
    int i, j, temp;
```